

CURRICULUM VITA

Yohn Jairo Parra Bautista, Ph.D.
Computational & Data-Enabled Science & Engineering [Big Data Science]
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PERSONAL
Yohn Jairo Parra-Bautista, Ph.D.
Data Science and Engineering
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GOAL: Engage in research and teaching in an academic institution as a knowledge creation researcher using data science, engineering, and analytics to provide: 1. The ability to solve societal/industrial problems by equipping decision-makers with the knowledge that can assist in the mitigation of situations as well as their effects on society; 2. By embracing new technologies and experiential learning, this generation of students can gain significant opportunities in this new and exciting field.

ACADEMIC

August 2023 -Present, Assistant Professor(tenure-earning) at Florida Agricultural and Mechanical University, Department of Computer and Information Sciences. Teaching and assisting in the development and implementation of the curriculum in Data Science. I had taught courses like Python, Databases, Ethics and Data Mining which I design it aligning data science principles.

August 2019 -May 2023, Visiting Assistant Professor at Florida Agricultural and Mechanical University, Department of Computer and Information Sciences. Teaching and assisting in the development and implementation of the curriculum in Data Science. I had taught courses like Python, Databases, Ethics and Data Mining which I design it aligning data science principles.

PROFESSIONAL ACTIVITIES

1. Microsoft Visiting Data Science Educator Grant, May 3 – May 28, 2021. I collaborate with a Data Science team, attend meetings, gather knowledge from the industry perspective, and share ideas. Responsible Artificial Intelligence, Machine Learning, and Differential Privacy are areas that Microsoft leads with inclusiveness and impact in all their products.

2. Curriculum Development Workgroups Issued by NSF National Data Science Alliance
The earner of the Curriculum Development Workgroups badge met bi-monthly over the academic year on an institutional team, along with other HBCU faculty staff, to develop a proposal for a new data science major program. 2023-2024

3. Campus advisor for the FAMU students in the Mastercard Data Challenge 2024- *The Team called “The Solution” secure second place in the competition.*

4. Summer sections I 2022-Present, Data Science abroad program with JCU university in Rome, Italy. *I work on creating a curriculum for the data science course we teach in JCU university every summer since 2021. We have students from FAMU granting scholarships from Microsoft that have a broad experience while learning with international students.*

5. Writing Across the Curriculum Summer Learning Community, May 24- June 18, 2021. I am participating in a series of workshops and training that will help me to transform my teaching practices and aid students in enhancing their writing and communications skills as students and young professionals. Funded by FAMU.

6. REU in Data Science and A.I. Across the NSF Florida -Georgia Louis Stokes Alliance for Minority Participation, June 7 – July 31, 2021, to present. This REU requires our creating an entire curriculum as this is a new field for which we wish to engage undergraduates. We will engage and challenge students in the data

science research field with lectures and speakers related to topics in Machine Learning, Deep Learning and A.I. Students will learn how to create a data science methodology, and they will present a research paper at the end of the REU. Lectures and Labs will be recorded and available 24/7. Funded by the NSF FG-LSAMP.

7. Infusing Data Science and AI in the Undergraduate Curriculum. We are creating, teaching and implementing this new College of Science and Technology Program to engage FAMU undergraduates in Data Science and AI tools. This is the CST BS+ program in which competitively selected undergraduates from various disciplines are engaged in cohorts receiving stipends to be engaged in one additional course per semester [Python, R, Machine Learning etc.] and in summers be engaged in faculty led research. We are creating, implementing, and teaching the material. Funded by NSF FG-LSAMP.

EDUCATION

September 2014- December 2019, Jackson State University, Jackson, MS, College of Science, Engineering and Technology Ph.D. Program- Computational and Data-Enabled Science & Engineering (Big Data Science, Engineering and Analytics) [94 semester hours] Awarded: May 3, 2020

1. Dissertation; Sentiment Analysis of Consumer Complaints: An Examination of Lexicon-Based and Deep Learning Approaches: We utilize an unsupervised lexicons approach and supervise Naïve Bayes and Unsupervised Word2Vec to create a model help decision-makers understand better the relationships between financial services and predatory lending thematic. We interconnect three general- purpose lexicons and one domain-specific lexicon with the idea to train complaint data with those lexicons. Using data exploration with sentiment analysis we found ambiguity between those unigram words related with Predatory Lending Thematic (PLT) into the whole dataset. Word2Vec model as a deep learning approach allow find semantic candidate interchangeable words that match the PLT to create a new corpus to train the model.
2. Research Interests: Text and Unstructured Data Mining and Analysis, Sentiment Analysis, Natural Language Processing, Artificial Intelligence [Machine and Deep Learning applications], Automated Decision Making with applications in Precision Medicine/ Health, Business Intelligence and Deep Learning applications; Global Supply Chain/Logistics
3. Research Tools: RStudio and Python (programming software), Endnote, Hadoop architecture, Spark cluster, SQL, Amazon Web Services (AWS), Tableau, XSEDE, HPC
4. Webinars and Podcasts Attended:
 - a. Critical Thinking in Data Science (with Debbie Berebichez),
 - b. Patterns for Successful Data Science Projects (Data Science Central, March 14, 2019),
 - c. Clean Data & Accurate Machine Learning Models (Data Science Central, February 18, 2019),
 - d. Creating Business Applications with R & Python (DSC, February 18, 2019),
 - e. A.I. Models and Active Learning (DSC, December 4, 2018),
 - f. State of the Art Deep Learning on Apache Spark (DSC, October 31, 2018)

Supplemental Instructions:

- a. Dataquest, provides hands-on training with data science projects
- b. Data camp, provides interactive courses on different thematics related to Big Data
- c. Edx, provides online courses with different universities that teach the fundamental knowledge of computer science, math and statistics

- d. Coursera provides online courses with content across many fields.
- e. Certifications in following [undertaken in collaborative learning seminars at Jackson State University under supervision of and examination by professor of record]
 - i. Edx, Data Science: Productive Tools (Harvard University's PH125.5x)
 - ii. Edx, Natural Language Processing, (Microsoft DEV288x)
 - iii. Edx, Deep Learning with TensorFlow, (IBM: DLO120EN)

January 2014–August 2014, Jackson State University English Language Institute

January 2013 - December 2013, Specialization in Logistics Universidad EAFIT, Medellin, Colombia

September 2000- May 2006 Bachelor of Science in International Business (with honors) Universidad Santo Tomas, Bucaramanga, Colombia [210 semester hours]

INTERNSHIPS and GRADUATE ASSISTANTSHIPS

January 2017- December 2017: College of Science, Engineering and Technology, Jackson State University, Jackson State University [JSU]:

1. Internship: Big Data Analysis for Neurodegenerative Diseases with CSET faculty and Florida International University, Center for Advanced Technology and Education, funded by NSF SCOPE [Scalable Omnipresent Environment] grant [see below], Supervisors: Richard A. Aló PhD, JSU. Malek Adjouadi PhD, FIU
 - a. Utilizing the Alzheimer ADNI [Alzheimer's Disease Neuro Imaging] data for:
 - ï Exploring the social determinants in patients with some antecedent in cognitive impairment using brain images and clinical data.
2. Internship: GIS Projects, funded by NSF SCOPE grant [see below]
 - a. Using GIS tools to study at zip code level with tapestry data the social determinants that are high correlate with Alzheimer disease.

January 2016 – December 2016: College of Science, Engineering and Technology, Jackson State University:

3. Internship sponsored by INDRA Technologies, Barcelona, Spain, utilizing their Smart Cities designated Sofia Software Platform for:
 - a. Developing a Smart Parking System on campus using GoPro Cameras to capture images to be displayed on a platform (mobile, app, website) that can be used with a language program that Sofia provides to gain insights/ create knowledge for decision makers.
 - b. Sofia is a middleware that allows the interoperability of multiple systems and devices, offering a semantic platform to make real world information available to smart applications (Internet of Things)
4. Graduate Assistantship: National Science Foundation sponsored graduate assistantship to work on developing applications for:
 - a. NSF SCOPE grant [SCalable OminPresent Environment] with California Institute of Telecommunications and Information Technology [CalIT2] Falko Kuenster PI, U.C. San Diego; Subcontract to R. A. Alo; Major Instrumentation Grant which constructs and utilizes a 3D visualization wall to provide visualization capabilities and velocity for Big Data Volume
 - b. NSF SENSEI Grant [SENSor Environment Imaging] with University of Illinois Chicago, University of California San Diego, CalIT2, Scripps Institute of Technology and University of Hawaii; NSF Major Instrumentation Grant, which is developing spherical sensor camera for marine science and bio medical applications.

January 2015 – December 2015:

1. Internship: Toyota Haley Barbour Internship, Toyota Corporation [SEP]America: a.

Projects:

- i. Developed an automated Geospatial tool that generates potential Toyota customer hot spot segments by integrating the ESRI Tapestry data to be used by Toyota.
- ii. Developed visualization tool (using Geospatial Technologies) focused on end-user

SKILLS

- ï Native language: Spanish-Reading, Writing, and Speaking Proficiency excellent
- ï Language: English: Reading, Writing, Speaking Proficiency-excellent
- ï Computer Tools: Microsoft Office, Data Science/Analytics Software (SAP, R, Python, Tableau, Matlab), GIS Technologies, their integration into Data Analytics, Visualization Wall [Scalable Omni-Presence Environment- a collaborative visualization instrument for exploration of Big Data enabling users to Visualize B.D. from Nano to Micro to Mega scale]; XSEDE, Hadoop and Spark architectures

MEMBERSHIPS

- ï IEEE Computer Society
- ï Association for Computing Machinery (ACM)

EXTRA-CURRICULAR INTERESTS

Hiking, Gym, Meditation, Music, Physical Fitness, Poetry, Dance, Art, Writing, and Photography

PROFESSIONAL MEETINGS/ DEVELOPMENT/AWARDS

- ï Award: Hanover Writing Academy 2023-Professional development in writing grants to external agencies.
- ï Workshop: NSF San Diego Supercomputer Center Summer Institute, August 5 to August 9, 2019: week-long workshop focused on a broad spectrum of introductory-to-intermediate topics in High Performance Computing and Data Science for researchers in academia and industry, especially in domains not traditionally engaged in supercomputing, who have problems that cannot typically be solved using local computing resources. SDSC is providing resources to a larger number of modest-sized computational research projects that represent, in aggregate, a tremendous amount of scientific progress. The Summer Institute will give the attendees an overview of topics in High Performance Computing and Data Science and accelerate their learning process through highly interactive classes with hands-on tutorials on the Comet [PetaScale] Supercomputer. <https://si19.sdsc.edu/>
- ï Workshop: NSF Cyber Carpentry: Data Life-Cycle Training with DataNet Federation Consortium, July15 to July 26, University of North Carolina at Chapel Hill. The overarching goal of the Cyber Training program is to prepare, nurture, and grow the national scientific workforce for creating, utilizing, and supporting advanced cyberinfrastructure (CI) that enables cutting-edge science and engineering and contributes to the nation's overall economic competitiveness and security. <http://cybercarpentry.web.unc.edu/>
- ï Summit: Association of Computing Machinery-Institute of Mathematical Statistics Interdisciplinary Summit on the Foundations of Data Science, June 2019, San Francisco, CA-distinguished speakers and panelists addressing topics such as deep learning, reinforcement learning, fairness, and ethics, in addition to discussions about the future of data science and the role of ACM and IMS.

<https://www.acm.org/articles/bulletins/2019/march/acm-ims-data-science-summit-2019>

- ï Award: NSF INCLUDES ASSIST Grant through Great Minds, in STEM, October 6-9, 2018, Early-Career Faculty Symposium delivering faculty development workshops on the tenure process, NSF proposal and reviewer opportunities, effective mentorship, inclusive pedagogy, innovating practices from research and teaching, and best practices for engaging Underrepresented Minority Students; engagement with the larger HENAAC Conference audience and serve as judges for the Research Posters Competition.
- ï Award: NSF INCLUDES ASSIST Grant through Great Minds, in STEM October 8-11, 2017- agenda same as above.
- ï Conference: ACM Richard Tapia, Celebration of Diversity in Computing Conference, September 2017
- ï Conference: 15th LACCEI International Multi-Conference for Engineering, Education, and Technology, July 2017
- ï Award: NSF INCLUDES ASSIST Grant through Great Minds, in STEM, October 6-9, 2016, agenda same as above.
- ï Workshop Award: Global Environment for Network Innovations [GENI] Regional Workshop and GENI Engineering Conference (GEC) 25, March 13-17, 2017, Florida International University: introduction to GENI testbed and keynote speech plus two tutorials: My First GENI Experiment [SDN using OpenFlow], and an advanced track (Wireless, Distributed Computing using Hadoop).
- ï Award: Texas Advanced Computational Center Summer Supercomputing Institute: High Performance Computing Training, Austin, Texas, August 1-5, 2016: introduced researchers on high performance computing, data analytics, and scientific visualization; how to effectively use advanced TACC computing resources and technologies for big data analysis and scientific visualization of their research.
- ï Conference: Supercomputing 2015, November 2015, Austin, Texas
- ï Conference: Great Minds in STEM October 2015 Pasadena
- ï Workshop: NSF XSEDE Workshop, New Orleans, September 8-9, 2015: training as Graduate Student Champion for NSF HPC resources
- ï Conference: 14th international Symposium on Recent Advances in Environmental Health Research, September 2016, Jackson State University.
- ï Workshop: Logistics in Supply Chain Management, SAP software Logistics, SAP University Annual Conference, Las Vegas, May 2015

PROFESSIONAL PUBLICATIONS / PRESENTATIONS

Time Series Analysis of Cryptocurrency Prices Using Long Short-Term Memory, *JP Fleischer, G von Laszewski, C Theran, YJ Parra Bautista Algorithms*, 15(7), 2022.

Health Disparities Through Generative AI Models: A Comparison Study Using a Domain

Specific Large Language Model

YJ Parra Bautista, C Theran, R Aló, V Lima

Proceedings of the Future Technologies Conference, pp. 220-232, 2023.

Time Series Analysis of Blockchain-Based Cryptocurrency Price Changes

J Fleischer, G von Laszewski, C Theran, YJ Parra Bautista

arXiv preprint arXiv:2202.13874, 2022.

DeepTrace: Improving Pandemic Health Care by Identifying Disparities and Determinants

J Liu, R A. Alo, YJ Parra Bautista

2021 IEEE International Conference on Big Data and Smart Computing (BigComp), pp. 0-4, 2021.

DeepTrace: Improving US Pandemic Health Care Through Health Disparity Identification and Determinant Tracing

J Liu, R A. Alo, YJ Parra Bautista

2020 International Conference on Intelligent Data Science Technologies and Applications (IDSTA)

Marital Status of Never Married with Rey Auditory Verbal Learning Test Cognition Performance is Associated with Mild Cognitive Impairment

YJ Parra Bautista, SS Messeha, C Theran, R Aló, C Yedjou, V Adankai, et al.

Applied Sciences, 13(3), Article 1656, 2023.

Behavior Analysis of Pandemic Source Media Communications

YJ Parra Bautista, J Liu, R A. Alo

2021 IEEE International Conference on Big Data and Smart Computing (BigComp), 2021.

Comparison of Lexicon Performances on Unstructured Behavioral Data

YJ Parra Bautista, R A. Alo, D Olson

2019 Sixth International Conference on Social Networks Analysis, Management and Security (SNAMS), 2019.

Ethical Considerations of Generative AI: A Survey Exploring the Role of Decision Makers in the Loop

YJ Parra Bautista, C Theran, R A. Alo

Proceedings of the AAAI Symposium Series, 3(1), pp. 391-398, 2024.

A Geospatial and ML-based Approach to Health Disparity Identification and Determinant Tracing for Improving Pandemic Health Care, J Liu, R A. Alo, YJ Parra Bautista, C G. Yedjou, C Theran, 2021 Eighth International Conference on Social Network Analysis, Management and Security (SNAMS), 2021

Deep Learning, Cloud Computing for Credit/Debit Industry Analysis of Consumer Behavior Parra Bautista, R A. Alo, N N. Wang 2020 7th IEEE International Conference on Cyber Security and Cloud Computing (CSCloud), 2020.

Machine Learning Computational Framework for Alzheimer's Disease Stages Classification

C Theran-Suarez, YJ Parra Bautista, V Adankai, R A. Alo

Proceedings of the Future Technologies Conference, pp. 381-397, 2022.

Data Science and Deep Learning Across the Alliance: A Personalized Project Activated

Modality REU, *YJ Parra Bautista, C Theran, R A. Alo*, 2022 7th International STEM Education Conference (iSTEM-Ed), pp. 1-4, 2022.

Health Disparities Through Generative AI Models: A Comparison Study Using a Domain Specific Large Language Model

Y Jairo Parra Bautista, V Lima, C Theran, R Alo
arXiv e-prints, arXiv:2310.18355, 2023.

Proceedings 2021 FAMU REU Supported by Cybertraining-DSC, *B G. Gregor von Laszewski, Yohn Jairo Parra Bautista, Carlos Theran, Geoffrey, et al.* Cybertraining-DSC, 12, 2021.

Chemo-Preventive Effect of Vegetables and Fruits Consumption on the COVID-19 Pandemic

C G. Yedjou, R A. Alo, J Liu, J Enow, P Henry, L L. P. B. T. Clement, et al.
Henry, 7, 2021.

Behavior Analysis of Pandemic Source Media Communications

YJ Parra Bautista, J Liu, R A. Alo
2021 IEEE International Conference on Big Data and Smart Computing (BigComp).

ChatGPT vs. Latimer Chatbox: An Experimental Comparison of Jailbreak Constraints in Romantic Fraud Context (Accepted)

YJ Parra Bautista, Hongmei Chi, Sharif Abdul, Vinicius Lima
4th IEEE International Conference on AI in Cybersecurity (ICAIC) (Online and In-Person Conference), 5-7 February 2025

SYNERGISTIC ACTIVITIES

1. Involved in developing an automated Geospatial tool that generates potential Toyota customer hot spot segments by integrating the ESRI Tapestry data that can be used by Toyota.
2. Worked on a project involving visualization of End-use commodity data about Exports of Goods by principal End-use Category in USA category using Geospatial technologies.
3. Microsoft Visiting Data Science Educator Grant, May 3 – May 28, 2021. I collaborate with a Data Science team, attend meetings, gather knowledge from the industry perspective, and share ideas. Responsible Artificial Intelligence, Machine Learning, and Differential Privacy are areas that Microsoft leads with inclusiveness and impact in all their products.

COLLABORATORS & OTHER AFFILIATIONS: Toyota Haley-Barbour Internship, Spatial Analysis to Model Market Segmentation Using GIS tools, June 2015

GRADUATE ADVISORS AND DOCTORAL STUDIES SPONSORS.

1. Richard Alo, PhD, Dean, College of Science and Technology, Florida A & M University.
2. Sudha Yerramilli, PhD, Interim Director, National Center for Biodefense Communications, Jackson State University
3. David Olson, PhD, James & H.K. Stuart Chancellor's Distinguished Chair Department of Supply Chain Management/ Analytics College of Business, University of Nebraska- Linc
4. NingNing Wang, PhD, Assistant Professor, Math and Statistical Sciences, Jackson State University
7. Sungbum Hong, PhD, Associate Professor, Computer Science, Jackson State University
6. Kamal Ali, PhD, Chair of Computer Engineering, CSET, Jackson State University

